

Joy J. Bayan

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Research Fields: Macroeconomics; Monetary Economics; International Economics; Macroeconometrics

Education

Ph.D. in Economics, Indiana University (expected May 2027)	2021-2027
Graduate Studies in Mathematical Economics, Panthéon-Sorbonne University, Paris, France	2019-2020
B.A. in Global Studies/ M.A. in Public Policy, University of Minnesota	2012-2017

Research

Job Market Paper

A Heterogeneous-Beliefs Phillips Curve: Evidence from Asymmetric Inflation Expectations

Abstract: The standard New Keynesian Phillips Curve summarizes heterogeneous inflation expectations by their mean. The paper shows that the assumption fails when beliefs are heterogeneously distributed during crisis periods, but the relevant distributional features vary between macroeconomic regimes and geographic regions. Using regional panel data from the University of Michigan Survey of Consumers, I find that the asymmetric component of inflation expectations is associated with higher regional inflation in both the Great Recession and post-COVID periods, but the direction of the asymmetry differs across episodes. A leftward shift in the distribution is associated with higher inflation during 2008–2012, consistent with concerns about persistently low inflation, whereas a rightward shift is associated with higher inflation during 2020–2025, consistent with inflation-expectation de-anchoring. The post-COVID rightward shift in inflation expectations is driven by metro areas across all regions except North Central.

Presentations: Federal Reserve Bank of Philadelphia Research Seminar (2026), Federal Reserve Bank of Cleveland RISE conference (2026), Southern Economic Association (2026), Lewis University (2026)

Working Papers

Heterogeneous Grocery Price Inflation: Online Retail Transformation and Monetary Policy

with Xudong An (Federal Reserve Bank of Philadelphia)

and Rodney Ramcharan (University of Southern California)

Abstract: Using 100 million retailer transactions from 228,519 U.S. households in novel Numerator data (2019–2023), we show that low-income households experienced 0.6–1.3% more grocery price inflation than high-income households by December 2023. After 2022, the prices of fresh produce rose 5.3% more for low-income households. We identify three mechanisms that underlie these differences. First, greater adoption of online shopping eliminated geographic pricing advantages, reducing relative savings for low-income households by 4.4%. Second, the expansion of warehouse clubs increased the price premiums for low-income households by 6.4%, consistent with the constraints on bulk purchasing. Third, monetary policy tightening increased grocery prices for low-income households by 0.11% more, with larger effects in concentrated markets where these households disproportionately shop, particularly through Walmart.

Presentations: Federal Reserve Bank of Philadelphia (2026), Insurance Risk Economic Series (2027)

“From Regional to Aggregate: Trade Openness and the Phillips Curve Break After 2000”

Abstract: This paper examines whether trade openness can explain the break in the U.S. Phillips curve after 2000. Within an open-economy New Keynesian framework, US regional panel data are used, and China’s accession to the World Trade Organization in 2001 is exploited as a source of trade-induced structural change. I construct a shift-share instrument using pre-2000 regional employment exposure to tradable industries and changes in national trade openness to identify the effect of trade exposure on the relationship between inflation and the unemployment gap. I find that greater trade exposure flattens the regional Phillips curve: inflation responded weakly to changes in the unemployment gap in regions more exposed to trade and contributed to the flattening of the U.S. aggregate Phillips curve after 2000.

Presentations: Midwest Economic Association (2026), American Economic Association (2027)

Work in progress

Monetary Policy, Oil Shocks, and the Distribution of S&P 500 Stock Market Returns

This paper studies how oil supply, global demand, oil-specific demand, and monetary policy shocks affect the *distribution* of U.S. stock market returns. Moving beyond mean-based analyzes, I model the entire cross-sectional distribution of S&P 500 returns using functional data analysis. Functional principal components (FPCA) extracted from estimated return densities are embedded in a mixed vector autoregressive framework combined with a Kilian-style SVAR using monthly data from 1988–2018, and show that oil demand and monetary policy shocks transmit not through just average returns.

Presentations: Cleveland State University (2025), Indiana University (2024)

Gas Price Expectations, News and the Skewness of Inflation Expectations after 2022

Abstract: This paper shows—using household survey information from the University of Michigan Survey of Consumers—that gasoline-related news and belief move the skewness of the cross-section of one-year-ahead inflation expectations after 2022. Monthly survey measures of gas price expectations and news about gas from respondents were constructed. Gas expectations and news co-move with the skewness component of the expectations distribution; meanwhile, right-tail mass is highly persistent. Skewness also predicts the three-months of household expectations.

Research Experience

Federal Reserve Bank of Philadelphia

06/2026–present

Management Trainee, Ph.D. Research Intern, Retail Supervisory Modeling Team, SRC

- Analyzed 100 million household transaction records from Numerator panel data (2019–2023) to construct income-quartile-specific price indices and study heterogeneous inflation experiences.

Federal Reserve Bank of San Francisco

Summer 2024

Ph.D. Research Intern, Supervision, Risk and Credit Department

- Designed an automated SQL pipeline processing millions of commercial real estate (CRE) loan records from regional banks and developed a PCA-based framework to identify CRE risk factors.

Research Assistantships, Indiana University

2022–2024

Research Assistant to Professors Yoosoon Chang, Todd Walker, and Mostafa Beshkar

- Conducted research in functional data analysis, proxy vector autoregressions, macroeconomic modeling, and international investment disputes using MATLAB, Julia, and R.

Teaching Experience

Instructor: Fundamentals of Macroeconomics for Business (60 students), Indiana University Fall 2024

- Instructor of record for 60 students, responsible for lectures, exams, grading, and office hours.

Teaching Assistant: Introductory Statistics (Honors), Psychology Statistics, Gender Economics,

Economics of Race, Fundamentals of Macroeconomics for Business, Indiana University 2021–2025

Academic Service Experience

Accessibility Coordinator, Economics Department, Indiana University

2025–2026

- Coordinated accessibility accommodations for students and advised faculty on accessibility

Development Coordinator, Graduate Student Government, Indiana University

2023–2024

- Organized professional development workshops and social events for 10,000+ graduate students

Professional Experience

Minnesota Department of Health

2020–2021

Senior Research Analyst, COVID-19 Surveillance

- Developed metrics for Governor’s epidemiological daily briefings for statewide public health response

Minnesota Department of Revenue

2017–2019

Star of the North Fellow, Management Analyst 3, Income and Property Tax Divisions

- Developed automated reporting tools for legislative and fiscal inquiries from 3 million + records

Minnesota State Demographic Center

2017–2018

Star of the North Fellow, Research Analyst, Census 2020 Preparation

- Published 2020 population forecast adopted by Minnesota Legislature for congressional redistricting

Conferences and Presentations

2027:

American Economic Association — Chair, “Inflation and Expectations” session;

American Economic Association — Presenter, “Trade Policy” session

2026:

Federal Reserve Bank of Philadelphia;

Federal Reserve Bank of Cleveland;

Southern Economic Association;

Midwest Economic Association;

Illinois Economic Association; Lewis University

2025:

Cleveland State University; Indiana University

Fellowships & Awards

Experiential Learning Scholarship, IU Walter Center (Summer 2026, Fall 2026)

IU Daniel J. Duesterberg Travel Award(2025, 2026)

IU Economics Enrichment Travel Fund(2025, 2026)

IU Global Citizens Exchange Scholarship, O’Neill School (2024)

Summer Award, Center for Applied Economics and Policy Research 2023

Star of the North Fellowship, State of Minnesota (2017–2018)

Lloyd Short Award for Best Master’s Thesis, University of Minnesota, 2017

Professional Development

2026: Federal Reserve Bank of Cleveland, RISE workshop: Regime-Switching and Nonlinear Modeling

2026: Erdos Institute, Data Science Bootcamp

2025: Economic Machine Learning Lab Quant Conference, Indiana University

2025: Workshop on Estimation of Structural Models, University of Notre Dame

2024: Theory and Experiments in Monetary Economics, Indiana University

2024: Heterogeneous Agent Asset Pricing Workshop, Cleveland State University

2023: Advances in Econometrics, Indiana University

Technical Skills

Programming: R, Python, Stata, MATLAB, Julia, SQL, Dynare, Fortran

Methods: Non-parametric Estimation, Time Series, Panel Data, Structural VAR, Machine Learning

Data tools: Git/GitHub, L^AT_EX, Tableau, R Shiny, SQL Server, Census/BLS/FRED APIs

Languages: English, Russian, Mongolian; French (intermediate), German (intermediate)

References

Gerhard Glomm

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Xudong An

Vice President of Supervision, Risk and Credit Department

Federal Reserve Bank of Philadelphia

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Rodney Ramcharan

Professor of Finance and Business Economics

Marshall School, University of Southern California

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